Gayatri Vidya Parishad College of Engineering Autonomous

Electrical and Electronics Engineering | Course Code: EE201

Electrical Circuit Analysis - Electrical Circuit Theorems & Transient Respo

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Overview: Thevenin, Norton, Superposition, Maximum power transfer theorems, Laplace transform c

I. UNIT-WISE SYLLABUS BREAKDOWN & HIGH-YIELD TOPICS

Unit 1: Network Theorems DC & AC

Superposition, Thevenin, Norton, Maximum Power Transfer, Reciprocity, and Millman theorems with depe

Unit 2: DC & AC Transient Analysis

First-order RL and RC differential equations, Time constants, Second-order RLC series and parallel t

Unit 3: Laplace Transform in Circuit Analysis

Transformed impedance models of R, L, and C, Initial and final value theorems, Step, ramp, and impul

Unit 4: Two-Port Network Parameters

Impedance Z, Admittance Y, Hybrid H, and Transmission ABCD parameters, Interconnection of two-port n

II. HIGH-YIELD SEMESTER EXAMINATION QUESTIONS

1. Find the Thevenin equivalent circuit across specified terminals containing both independent and voltag
2. Solve for the transient current it in an underdamped series RLC circuit excited by a DC step voltage u
3. Derive the relationship between Z-parameters and ABCD transmission parameters for a reciprocal two-por

III. PRESCRIBED TEXTBOOKS & ACADEMIC REPOSITORIES

* William H. Hayt, Jack E. Kemmerly - Engineering Circuit Analysis McGraw-Hill 8th Ed
* Charles K. Alexander, Matthew N.O. Sadiku - Fundamentals of Electric Circuits McGraw-Hill

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